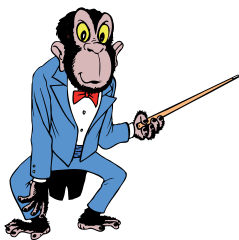




Solving Quadratics by Formula



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- A). Solve, with the above formula, the quadratics which are in the form $ax^2 + bx + c$. These quadratics will factorise so you can use this as a check.

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|----------------------------|--------------------------|----------------------------|
| 1). $x^2 + 5x + 6 = 0$ | 2). $x^2 - 5x + 4 = 0$ | 3). $x^2 - 4x - 5 = 0$ |
| 4). $x^2 - 6x + 5 = 0$ | 5). $x^2 + 2x - 8 = 0$ | 6). $x^2 - 10x + 21 = 0$ |
| 7). $2x^2 + 5x + 3 = 0$ | 8). $2x^2 + 5x - 3 = 0$ | 9). $3x^2 - 4x + 1 = 0$ |
| 10). $3x^2 - 5x - 2 = 0$ | 11). $6x^2 + 7x + 2 = 0$ | 12). $3x^2 - 13x - 10 = 0$ |
| 13). $6x^2 - 11x - 10 = 0$ | 14). $5x^2 - 3x - 2 = 0$ | 15). $4x^2 + 7x - 2 = 0$ |
| 16). $6x^2 + 13x + 6 = 0$ | 17). $5x^2 + 6x + 1 = 0$ | 18). $8x^2 - 6x + 1 = 0$ |

- B). Solve the following quadratics, using the above formula to 2 decimal places.

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|--------------------------|---------------------------|---------------------------|
| 1). $x^2 + 6x + 4 = 0$ | 2). $x^2 + 5x + 3 = 0$ | 3). $x^2 + 2x - 5 = 0$ |
| 4). $x^2 - 10x + 8 = 0$ | 5). $x^2 + 12x + 10 = 0$ | 6). $x^2 + 7x + 4 = 0$ |
| 7). $2x^2 - 5x - 4 = 0$ | 8). $2x^2 + 9x + 3 = 0$ | 9). $3x^2 + 7x + 3 = 0$ |
| 10). $2x^2 - 3x - 1 = 0$ | 11). $5x^2 + 9x + 2 = 0$ | 12). $3x^2 + 2x - 3 = 0$ |
| 13). $5x^2 + x - 2 = 0$ | 14). $5x^2 + 8x + 2 = 0$ | 15). $2x^2 - 11x + 7 = 0$ |
| 16). $3y^2 + 6y - 7 = 0$ | 17). $4p^2 + 7p - 6 = 0$ | 18). $5a^2 + 9a + 2 = 0$ |
| 19). $b^2 + 2b - 5 = 0$ | 20). $q^2 - 15q + 8 = 0$ | 21). $2t^2 - t - 4 = 0$ |
| 22). $3w^2 - 5w - 8 = 0$ | 23). $2x^2 + 11x + 8 = 0$ | 24). $3r^2 + 8r + 3 = 0$ |
| 25). $5k^2 + 9k + 2 = 0$ | 26). $x^2 + 3x + 1 = 0$ | 27). $x^2 - 2x - 4 = 0$ |
| 28). $2x^2 + 7x - 3 = 0$ | 29). $3x^2 - 5x - 3 = 0$ | 30). $5x^2 + 3x - 3 = 0$ |

- C). Simplify and solve the following quadratics to 2 decimal places.

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|----------------------|-----------------------|------------------------|
| 1). $x^2 + 10x = 5$ | 2). $2x^2 + 5 = 9x$ | 3). $2k^2 + 4k = 3$ |
| 4). $4c^2 + 9c = 3$ | 5). $2x^2 - x = 7$ | 6). $x^2 + 6 = 8x$ |
| 7). $4x^2 + 3x = 5$ | 8). $5x^2 = 7x - 1$ | 9). $3x^2 + 5 = 9x$ |
| 10). $3x^2 = 7x - 3$ | 11). $x^2 - 8x = 7$ | 12). $x^2 + 5x = 2$ |
| 13). $2x^2 = 7x + 3$ | 14). $4x^2 - 9x = 3$ | 15). $x^2 = 8x - 11$ |
| 16). $5x^2 - 2 = 6x$ | 17). $5 - 7x = 2x^2$ | 18). $14x = 3x^2 + 8$ |
| 19). $6x + 3 = 5x^2$ | 20). $2 = 5x^2 + 8x$ | 21). $12x = 3x^2 + 10$ |
| 22). $8x = 3x^2 + 2$ | 23). $16x = 4x^2 + 3$ | 24). $2x^2 = 7x - 3$ |

- D). Simplify and solve the following quadratics to 2 decimal places.

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|------------------------------|------------------------------------|-------------------------------|
| 1). $x(x - 2) = 5$ | 2). $2x(x + 3) + 3 = 0$ | 3). $(x - 2)(x - 4) = 5$ |
| 4). $(x + 1)(x - 1) = 1$ | 5). $(x + 1)(x - 5) = 3$ | 6). $(2x + 1)(x - 1) = 1$ |
| 7). $(2x - 5)(x + 3) = 8$ | 8). $3x(x - 1) = 5$ | 9). $(x + 3)^2 = 2$ |
| 10). $2x(x + 4) = 1$ | 11). $3x(x - 2) = 2$ | 12). $x(x + 8) = 2x + 1$ |
| 13). $5x(x + 3) = 3(8x - 1)$ | 14). $(x + 3)(x + 1) = 4$ | 15). $(x - 3)(2x - 5) = 2$ |
| 16). $4 = 5x(4 - 5x)$ | 17). $(5x - 3)(3x + 1) = 1$ | 18). $5x(2x - 3) = 7(3 - 2x)$ |
| 19). $x(x + 4) = 6(x + 4)$ | 20). $2(2x + 1)^2 + 5(2x + 1) = 1$ | |

